

Coupled Hamiltonian Shear Degree Matrices in $\mathbb{A}^4$: An Asymmetric $g \geq 5$ Family

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Abstract

We prove an exact degree formula for a fixed family of canonical polynomial automorphisms of four-dimensional affine space. Let K be algebraically closed of characteristic zero, let $g \geq 5$, and compose the two triangular Hamiltonian shears generated by $V(q) = q_1^2 q_2^2 + q_1^g$ and $W(p) = p_1^2 p_2^2 + p_2^g$. The mixed supports force a genuinely coupled two-phase degree calculation. We show that the first shear selects the degree-transfer matrix $A_g = ((g-1, 0), (2, 1))$, the second selects $B_g = ((1, 2), (0, g-1))$, and the complete step is governed by $C_g = B_g A_g$, not by the off-diagonal matrix that records the two half-steps. A half-open ratio cone is invariant under C_g ; strict old-term gaps and positive coefficients in characteristic zero prevent cancellation at every iterate. The second q -coordinate sees the Perron class, so

$$\deg(F_g^n) = e_2^T C_g^n (1, 1)^T, \quad \lambda_1(F_g) = \rho(C_g) = (\sqrt{g} + 1)^2.$$

Each elementary shear has degree $g-1$, whereas $(\sqrt{g} + 1)^2 < (g-1)^2$. The phrase *non-product* is used only for the displayed coordinate support and this strict degree comparison; no universal non-conjugacy statement is claimed.

1 Introduction

Polynomial automorphisms admit degree-growth mechanisms that are invisible if one only records the degree of each factor. In the planar generalized Hénon setting, reduced-word structure leads to a degree product law; this is a useful baseline, not a theorem about every higher-dimensional composition (Friedland and Milnor, 1989; Arnold Mathematical Journal, 2024). In higher dimension, several growth regimes occur and algebraic stability requires hypotheses that must be checked rather than inferred from a formal matrix (Déserti, 2018; Guedj and Sibony, 2002; Favre and Jonsson, 2011; Dang and Favre, 2021). Our aim is to isolate one small family in which all of the checks can be made explicitly.

We study a composition of two canonical shears on $\mathbb{A}_K^4$, with $q = (q_1, q_2)$ and $p = (p_1, p_2)$. The potentials

$$V(q) = q_1^2 q_2^2 + q_1^g, \quad W(p) = p_1^2 p_2^2 + p_2^g$$

are asymmetric: one pure power occurs in the first q -gradient row and the other in the second p -gradient row. The mixed quartic monomials keep the two coordinates coupled. We fix the word $F_g = T \circ S$ before doing any degree analysis:

$$S(q, p) = (q, p + \nabla V(q)), \quad T(q, p) = (q + \nabla W(p), p).$$

This order matters. The first shear produces an intermediate p -degree vector, and the second shear consumes that vector before the complete iterate returns to a q -degree vector.

The central point is a phase-sensitive selector calculation. If $u = (u_1, u_2)^\top$ records the degrees of the q -coordinates, the selected rows of the first shear are

$$A_g = \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix}.$$

After the intermediate vector $v = A_g u$ is formed, the second shear selects

$$B_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix}.$$

The full return map is therefore

$$C_g = B_g A_g = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix}.$$

The matrix C_g , rather than a half-step matrix, is the object whose Perron root can control the degree sequence.

Three technical issues make the statement nonformal. First, the selected faces must dominate both competing monomials and the carried coordinate. This requires a cone in the q -phase and a separate comparison in the intermediate p -phase. Second, the cone must be invariant under the complete matrix for every $g \geq 5$, not just at one initial step. Third, a spectral radius is not automatically a total degree: one must prove that the initial vector reaches the positive Perron class and that a concrete coordinate functional sees it. We address these points by strict inequalities and an explicit visibility argument.

The paper makes four bounded, internally provable contributions:

1. It gives an explicit symplectic polynomial automorphism with triangular inverses and a connected mixed support in the displayed split.
2. It proves a two-phase Newton-face selector for every integer $g \geq 5$, including the carried-coordinate inequalities and an invariant ratio cone.
3. It proves coefficientwise no-cancellation and the exact complete-step recurrence $u_{n+1} = C_g u_n$, then identifies total degree through the q_2 -coordinate.
4. It obtains the closed form $\lambda_1(F_g) = (\sqrt{g} + 1)^2$ and compares it strictly with $\deg(S) \deg(T) = (g-1)^2$, without making a universal product or priority claim.

The nearest four-dimensional coupled-symplectic literature studies a different question, namely horseshoes and uniform hyperbolicity of coupled Hénon maps (Fujioka et al., 2023). We study algebraic degree and prove the relevant formula from the displayed supports. The remainder of the paper is organized around that proof: Section 2 fixes the bounded comparison boundary; Section 3 defines the family and phase notation; Sections 5–6 establish canonicity and face selection; Sections 7–8 prove the recurrence and its observable degree; and Sections 10–12 state the narrow non-product comparison, limitations, and conclusion.

2 Bounded related work and collision boundary

Planar degree products. The plane classification of polynomial automorphisms provides the classical degree-product intuition for reduced compositions (Friedland and Milnor, 1989). We use

that result only as a two-dimensional baseline. Our map lives in dimension four, is constrained to be canonical, and has a mixed support that produces a non-diagonal complete-step matrix. We do not infer a contradiction with the planar law and do not claim a classification of higher-dimensional degree growth.

Higher-dimensional degree growth. Examples in dimension at least three show that degree sequences need not fall into the simplest planar regimes (Déserti, 2018). General terminology for algebraic stability and projective degree growth is useful here, but it does not replace the face-by-face induction needed for this family (Guedj and Sibony, 2002). Valuation and compactification methods in dimension two provide historical context for degree growth (Favre and Jonsson, 2011); they are not invoked in the proof below.

Spectral language. Dynamical degrees are often interpreted through spectral radii under explicit stability or visibility hypotheses (Guedj and Sibony, 2002; Dang and Favre, 2021). In our setting the relevant operator is finite and elementary, but the same logical caution applies: the positive matrix alone is insufficient. We prove that its Perron class is reached from $(1, 1)^T$, and that e_2^T observes the maximal coordinate degree.

A four-dimensional neighbor. A coupled-Hénon literature in four dimensions studies symplectic coupling for topological horseshoes and uniform hyperbolicity (Fujioka et al., 2023). This is a neighboring object, not a collision: our potentials, degree-transfer matrices, cone, and closed-form algebraic degree are different, and no hyperbolicity or topological-entropy statement appears here.

2.1 Paper12–Paper19 boundary

The following table records the exact object boundary used in the bounded collision review. Each row is empty at that scope; the table is not an exhaustive priority search.

Ledger	Neighboring object	Paper20 boundary
P12	Residue/period-three arithmetic and clock strata	No residue classes, period-three data, or arithmetic clocks.
P13	Primitive exact-period covers and cyclic orbit quotients	No periodic points, orbit quotients, or dynatomic schemes.
P14	Support-one torus escape and finite-rank map geometry	No torus survivors, character lattices, or escape classification.
P15	Pure-trace/Jacobian fibers and quasi-finite loci	No trace coordinates, Jacobian fibers, or ramification.
P16	Support-size torus escape bounds	No support-size counting or finite-rank torus map.
P17	Fixed-lag shift-like recurrence and torus-coset dimension	No recurrence variety, lag word, character extinction, or coset dimension.
P18	Marked trace coordinates and scheme-theoretic boundary ramification	No trace marking, boundary scheme, or branch divisor.
P19	Maximum-dimensional translates, coefficientwise moduli, support-one GCD obstruction	No translates, GCD obstruction, effective height, or periodic classification.

The paper also excludes the prime-clock, cat-map, centralizer, zeta, and quantization lines from Papers1–11 by scope. If a later review identifies a neighboring result, the response is to narrow the claim and add a boundary, not to erase the comparison.

3 Fixed family, coordinates, and phase notation

Throughout, K is algebraically closed and has characteristic zero, and $g \geq 5$ is an integer. We work on $\mathbb{A}_K^4$ with coordinates (q_1, q_2, p_1, p_2) and standard symplectic form

$$\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2.$$

Set

$$V(q) = q_1^2 q_2^2 + q_1^g, \quad W(p) = p_1^2 p_2^2 + p_2^g.$$

The two triangular maps and their composition are

$$S(q, p) = (q, p + \nabla V(q)), \quad T(q, p) = (q + \nabla W(p), p), \quad F_g = T \circ S.$$

Writing $\hat{p} = p + \nabla V(q)$, the coordinates of one complete step are

$$\hat{p}_1 = p_1 + 2q_1 q_2^2 + g q_1^{g-1}, \quad \hat{p}_2 = p_2 + 2q_1^2 q_2, \quad (3.1)$$

$$F_{g,1} = q_1 + 2\hat{p}_1 \hat{p}_2^2, \quad F_{g,2} = q_2 + 2\hat{p}_1^2 \hat{p}_2 + g \hat{p}_2^{g-1}, \quad (3.2)$$

$$F_{g,3} = \hat{p}_1, \quad F_{g,4} = \hat{p}_2. \quad (3.3)$$

The formulas, rather than a formal Jacobian, determine the degree-transfer rows below.

For a polynomial h , $\deg h$ means total degree. For a polynomial map, $\deg F$ is the maximum of its coordinate degrees. Define

$$q_{i,n} = q_i \circ F_g^n, \quad p_{i,n} = p_i \circ F_g^n,$$

and set

$$u_n = \begin{pmatrix} \deg q_{1,n} \\ \deg q_{2,n} \end{pmatrix}.$$

The intermediate vector v_{n+1} records the p -degrees immediately after the S -phase applied to F_g^n , before the T -phase:

$$v_{n+1} = \begin{pmatrix} \deg(p_{1,n} + \partial_{q_1} V(q_n)) \\ \deg(p_{2,n} + \partial_{q_2} V(q_n)) \end{pmatrix}.$$

For the carried initial coordinates we set $u_0 = v_0 = (1, 1)^\top$. Thus v_n and u_n have different phase meanings even when their entries happen to have comparable size.

The four selected monomials and their degree rows are

phase row	selected monomial	degree in the input vector
$S, 1$	q_1^g	$(g-1)u_1$
$S, 2$	$q_1^2 q_2^2$	$2u_1 + u_2$
$T, 1$	$p_1^2 p_2^2$	$v_1 + 2v_2$
$T, 2$	p_2^g	$(g-1)v_2$

Consequently,

$$A_g = \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix}, \quad B_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix}.$$

These rows are Newton-face exponent rows for the displayed monomials. They are not free Jacobian data, and every use of them below is accompanied by a strict selector inequality.

Definition 3.1 (Two-phase cone). For $u = (u_1, u_2)^\top \in \mathbb{R}_{>0}^2$, write $r(u) = u_2/u_1$ and define

$$\mathcal{C}_g = \left\{ u \in \mathbb{R}_{>0}^2 : 1 \leq r(u) < \frac{g-2}{2} \right\}.$$

The q -phase is the cone $\mathcal{C}_g$; the intermediate p -phase is the image $A_g \mathcal{C}_g$, with its own selector inequalities.

Theorem 3.2 (Asymmetric coupled-shear degree matrix). For every integer $g \geq 5$, F_g is a symplectic polynomial automorphism of $\mathbb{A}_K^4$. If

$$C_g = B_g A_g = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix},$$

then, for every $n \geq 1$,

$$\deg(F_g^n) = e_2^\top C_g^n \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (3.4)$$

and

$$\lambda_1(F_g) = \lim_{n \rightarrow \infty} \deg(F_g^n)^{1/n} = \rho(C_g) = (\sqrt{g} + 1)^2. \quad (3.5)$$

Each elementary shear has degree $g-1$, so

$$\lambda_1(F_g) < (g-1)^2 = \deg(S) \deg(T).$$

The mixed support makes the displayed coordinate split coupled. No non-conjugacy statement is asserted for other coordinate systems.

4 Degree bookkeeping and support filtration

We use ordinary total degree, but it is helpful to separate the elementary properties of degree that make the selector argument rigorous. If $u = (u_1, u_2) \in \mathbb{R}_{\geq 0}^2$ and $m = q_1^{a_1} q_2^{a_2}$, define its u -weight by

$$\text{wt}_u(m) = a_1 u_1 + a_2 u_2.$$

For a polynomial in q_1, q_2 , its u -weighted degree is the maximum of the weights of its monomials with nonzero coefficient. Substitution of coordinate polynomials of degrees u_1, u_2 gives an upper bound by this weighted degree. If a monomial attaining the maximum has a nonzero coefficient and no competing summand has the same degree with an opposite coefficient, the upper bound is attained.

The only weighted-degree comparisons used in this paper are finite. For the first gradient row they are

$$\text{wt}_u(q_1 q_2^2) = u_1 + 2u_2, \quad \text{wt}_u(q_1^{g-1}) = (g-1)u_1.$$

For the second shear they are

$$\text{wt}_v(p_1 p_2^2) = v_1 + 2v_2, \quad \text{wt}_v(p_1^2 p_2) = 2v_1 + v_2, \quad \text{wt}_v(p_2^{g-1}) = (g-1)v_2.$$

The strict inequalities in Sections 6 and 7 say that the displayed maximum is unique at the level of degree. They are stronger than an assertion that a row is generically selected: they hold for every degree vector in the invariant cone.

phase	output	source kind	coefficient	exponent row	weighted degree
S	$\widehat{p}_1$	carry p_1	1	–	v_1
S	$\widehat{p}_1$	mixed gradient	2	$(1, 2)$	$u_1 + 2u_2$
S	$\widehat{p}_1$	pure gradient	g	$(g - 1, 0)$	$(g - 1)u_1$
S	$\widehat{p}_2$	carry p_2	1	–	v_2
S	$\widehat{p}_2$	mixed gradient	2	$(2, 1)$	$2u_1 + u_2$
T	q'_1	carry q_1	1	–	u_1
T	q'_1	mixed gradient	2	$(1, 2)$	$v_1 + 2v_2$
T	q'_2	carry q_2	1	–	u_2
T	q'_2	mixed gradient	2	$(2, 1)$	$2v_1 + v_2$
T	q'_2	pure gradient	g	$(0, g - 1)$	$(g - 1)v_2$

Table 1: Complete support and phase ledger.

Lemma 4.1 (Max-plus substitution bound). *Let $h = \sum_{\alpha} c_{\alpha} x^{\alpha}$ and let x_i be replaced by polynomials of degrees d_i . Then*

$$\deg h \leq \max_{\alpha: c_{\alpha} \neq 0} \sum_i \alpha_i d_i.$$

If one monomial on the right has a coefficient that remains nonzero and has strictly larger degree than all other summands in the same addition, equality holds.

Proof. The degree of a product is the sum of the degrees of its factors, and the degree of a sum is at most the maximum of the summand degrees. Applying these facts to each monomial and then taking the maximum proves the bound. A strict maximum cannot be cancelled by a lower-degree summand; if its coefficient is nonzero, it realizes the bound. $\square$

4.1 Complete support and phase ledger

We now make the finite support calculation explicit, including the terms that are carried through a triangular shear. Differentiation removes one exponent from a potential monomial; the remaining exponent vectors are the candidate rows acting on a degree vector. The potential supports themselves are

$$\text{supp}(V) = \{(2, 2), (g, 0)\}, \quad \text{supp}(W) = \{(2, 2), (0, g)\}.$$

After differentiation, the nonzero gradient supports and their coefficients are

$$\partial_{q_1} V = 2q_1 q_2^2 + g q_1^{g-1}, \quad \partial_{q_2} V = 2q_1^2 q_2, \quad (4.1)$$

$$\partial_{p_1} W = 2p_1 p_2^2, \quad \partial_{p_2} W = 2p_1^2 p_2 + g p_2^{g-1}. \quad (4.2)$$

There are no omitted derivative rows: every term in the two potentials appears in exactly the gradient coordinates predicted by its exponent. This finite fact is what allows us to use strict inequalities rather than a generic Newton-fan assertion.

For a q -degree vector u and an intermediate p -degree vector v , the complete ledgers are as follows. The *carry* column records the coordinate already present before the relevant shear; it must be compared separately with the gradient candidates. The table is an occurrence ledger, not an additional map definition. In the S rows u measures the old q coordinates and v measures the carried p coordinates; in the T rows the roles are reversed. The selected entries are the pure first-row term in S , the unique second-row term in S , the mixed first-row term in T , and the pure second-row term in T . This is exactly the row selection encoded by A_g and B_g .

For a formal support notation, let

$$\text{supp}_q(P) = \{\alpha \in \mathbb{N}^2 : [q^\alpha]P \neq 0\}, \quad \deg_u(P) = \max_{\alpha \in \text{supp}_q(P)} \langle \alpha, u \rangle.$$

The analogous supp_p and $\deg_v$ are used in the T phase. If P_i are substituted for q_i , a term indexed by α expands into products of terms from the P_i , and every such product has degree at most $\langle \alpha, u \rangle$. Thus Lemma 4.1 is a finite support statement: it takes the maximum of homogeneous linear forms, one for each support exponent. The carry rows in the table are not support exponents of a potential; they arise from the identity summand in the triangular map and are therefore tested by a separate old-term inequality.

The strict tests can be organized into a phase ledger. There are two different intermediate vectors in the S comparison: the carried vector from the preceding iterate and the newly produced vector. Write $\bar{v}$ for the carried p -degree vector and put $v^+ = A_g u$ for the new output of the S phase. (In the induction below, $\bar{v} = v_n = A_g u_{n-1}$ and $v^+ = v_{n+1} = A_g u_n$.) The selected-minus-competitor margins are

$$(g-1)u_1 - (u_1 + 2u_2) = u_1(g-2-2u_2/u_1), \quad (4.3)$$

$$(g-1)u_1 - \bar{v}_1 > 0, \quad (2u_1 + u_2) - \bar{v}_2 > 0, \quad (4.4)$$

$$(v_1^+ + 2v_2^+) - u_1 = (g+2)u_1 + 2u_2, \quad (4.5)$$

$$(g-1)v_2^+ - (2v_1^+ + v_2^+) = (g-2)u_2 - 2u_1, \quad (4.6)$$

$$(g-1)v_2^+ - u_2 = 2(g-1)u_1 + (g-2)u_2. \quad (4.7)$$

The first and fourth expressions are face comparisons; the remaining three compare a selected gradient contribution with an identity carry. Keeping these classes separate prevents a high-degree old coordinate from being silently discarded. The simultaneous induction in Proposition 7.3 proves the carry inequalities at every step, including the base step. Notice that substituting $v^+ = A_g u$ into the two S carry expressions would give zero identities, not strict inequalities; those expressions must therefore use the genuinely old vector $\bar{v}$.

Remark 4.2 (Why a face row is not a Jacobian row). The matrices A_g and B_g encode weighted degrees of four specific support monomials after differentiating the potentials. A Jacobian matrix records first derivatives of the full map and has a different meaning. In particular, replacing A_g or B_g by a Jacobian or Hessian matrix would discard the carried-coordinate comparisons and would not prove (6.6).

4.2 The phase filtration

The complete step has the filtration

$$\underbrace{u_n}_{q\text{-phase}} \xrightarrow{S} \underbrace{v_{n+1}}_{p\text{-phase}} \xrightarrow{T} \underbrace{u_{n+1}}_{q\text{-phase}}.$$

The first arrow is not an iteration of F_g , and the second arrow cannot be applied to u_n directly. This distinction is relevant even if one only wants the maximal degree: the inequalities for the S -phase involve the ratio u_2/u_1 , whereas the second T -row involves $v_2/v_1 = (2 + u_2/u_1)/(g-1)$. The latter ratio is not the former ratio.

The initial values are deliberately assigned to both phase ledgers: $u_0 = v_0 = (1, 1)^\top$. The value v_0 represents the original degree-one p -coordinates only for the base comparison. For every genuine step $n \rightarrow n+1$, v_{n+1} is defined by the S -phase and is then consumed by T . This convention lets the base case and the induction use the same displayed inequalities without silently identifying the phases.

4.3 A useful matrix comparison

The complete matrix is positive, but the phase matrices are only nonnegative:

$$A_g = \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix}, \quad B_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix}.$$

The off-diagonal stacked matrix

$$M_g = \begin{pmatrix} 0 & B_g \\ A_g & 0 \end{pmatrix}$$

satisfies

$$M_g^2 = \begin{pmatrix} B_g A_g & 0 \\ 0 & A_g B_g \end{pmatrix}.$$

Thus $\rho(M_g)^2 = \rho(B_g A_g)$. A statement about one application of M_g would therefore describe a half-step; the degree per complete iterate is controlled by $\rho(C_g)$, not by $\rho(M_g)$ without the square. This elementary observation prevents a common phase-counting error.

5 Canonical shears and explicit inverses

The canonicity calculation fixes the algebraic object before the degree argument begins. We first record the inverse maps in coordinates, and then verify the symplectic identity both by pullback and by the block Jacobian.

Lemma 5.1 (Triangular inverses). *The maps S and T are polynomial automorphisms with*

$$S^{-1}(q, p) = (q, p - \nabla V(q)), \quad T^{-1}(q, p) = (q - \nabla W(p), p).$$

Consequently $F_g^{-1} = S^{-1} \circ T^{-1}$ is polynomial. More explicitly, if

$$q_1^- := q_1 - 2p_1 p_2^2, \quad q_2^- := q_2 - 2p_1^2 p_2 - g p_2^{g-1},$$

then

$$\begin{aligned} F_g^{-1}(q, p) &= (q_1^-, q_2^-, p_1 - 2q_1^-(q_2^-)^2 - g(q_1^-)^{g-1}, \\ &\quad p_2 - 2(q_1^-)^2 q_2^-). \end{aligned} \tag{5.1}$$

Proof. The map S leaves q fixed, so subtracting $\nabla V(q)$ recovers p . The map T leaves p fixed, so subtracting $\nabla W(p)$ recovers q . This proves the two displayed triangular inverses by direct composition in both orders.

For the complete map, T^{-1} first produces

$$(q_1^-, q_2^-, p_1, p_2) = (q - \nabla W(p), p).$$

Applying S^{-1} then subtracts $\nabla V(q^-)$ from p , which gives (5.1). Every entry is polynomial, so this is an explicit polynomial inverse rather than an appeal to a Jacobian criterion. $\square$

Lemma 5.2 (Symplecticity and block Jacobian). *Both S and T , hence F_g , preserve $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$. In the coordinate order (q_1, q_2, p_1, p_2) , their Jacobians and the Jacobian of F_g have the block forms*

$$DS(q, p) = \begin{pmatrix} I_2 & 0 \\ Q & I_2 \end{pmatrix}, \quad DT(q, \hat{p}) = \begin{pmatrix} I_2 & P \\ 0 & I_2 \end{pmatrix}, \quad (5.2)$$

$$DF_g(q, p) = \begin{pmatrix} I_2 + PQ & P \\ Q & I_2 \end{pmatrix}, \quad Q = H_V(q), \quad P = H_W(\hat{p}). \quad (5.3)$$

Proof. Write

$$H_V(q) = D^2V(q) = \begin{pmatrix} 2q_2^2 + g(g-1)q_1^{g-2} & 4q_1q_2 \\ 4q_1q_2 & 2q_1^2 \end{pmatrix}$$

and

$$H_W(p) = D^2W(p) = \begin{pmatrix} 2p_2^2 & 4p_1p_2 \\ 4p_1p_2 & 2p_1^2 + g(g-1)p_2^{g-2} \end{pmatrix}.$$

Both matrices are symmetric. For the first shear,

$$\begin{aligned} S^*\omega &= \sum_{i=1}^2 dq_i \wedge d(p_i + \partial_{q_i}V(q)) \\ &= \omega + \sum_{i,j=1}^2 \partial_{q_iq_j}V(q) dq_i \wedge dq_j = \omega. \end{aligned}$$

The last equality follows by pairing the (i, j) and (j, i) terms: their coefficient is $\partial_{q_iq_j}V - \partial_{q_jq_i}V = 0$.

For the second shear,

$$\begin{aligned} T^*\omega &= \sum_{i=1}^2 d(q_i + \partial_{p_i}W(p)) \wedge dp_i \\ &= \omega + \sum_{i,j=1}^2 \partial_{p_ip_j}W(p) dp_j \wedge dp_i = \omega, \end{aligned}$$

again by Hessian symmetry. Therefore

$$F_g^*\omega = S^*(T^*\omega) = S^*\omega = \omega.$$

The same calculation in block notation makes the phase order explicit. Set

$$J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$DS^\top JDS = \begin{pmatrix} Q - Q^\top & I_2 \\ -I_2 & 0 \end{pmatrix}, \quad DT^\top JDT = \begin{pmatrix} 0 & I_2 \\ -I_2 & P^\top - P \end{pmatrix}.$$

These matrices equal J because Q and P are symmetric. Since $DF_g = DT(q, \hat{p})DS(q, p)$, multiplication of the two triangular blocks gives (5.3). If

$$M := DF_g = \begin{pmatrix} I_2 + PQ & P \\ Q & I_2 \end{pmatrix},$$

then a direct composite check is

$$M^T J M = \begin{pmatrix} (I_2 + QP)Q - Q(I_2 + PQ) & (I_2 + QP) - QP \\ PQ - (I_2 + PQ) & P - P \end{pmatrix} = J.$$

The block triangular factors also give $\det DS = \det DT = 1$, hence $\det DF_g = 1$. This determinant identity is only a consistency check; polynomial invertibility was established by the explicit inverse above. The block matrices certify canonicity, whereas the support monomials and strict degree gaps below certify degree growth.

For completeness, the inverse Jacobians have the same triangular shape:

$$DS^{-1}(q, p) = \begin{pmatrix} I_2 & 0 \\ -Q & I_2 \end{pmatrix}, \quad DT^{-1}(q, p) = \begin{pmatrix} I_2 & -P \\ 0 & I_2 \end{pmatrix}.$$

Here Q and P are evaluated at the source point of the corresponding inverse shear. Multiplying $DS^{-1}DT^{-1}$ in the reverse phase order recovers DF_g^{-1} , and multiplying either pair with the forward blocks gives the identity. Thus the inverse in Lemma 5.1 is reflected both in coordinates and in the Jacobian factorization; no constant-Jacobian criterion is being used as a substitute for the explicit polynomial inverse. □

Remark 5.3. Characteristic zero is not needed for the formal Hessian-symmetry identity. It is used later when the positive integer coefficients 2 and g are interpreted as nonzero coefficients in K . No selector or degree statement for other potentials is asserted here.

6 The two Newton-face selectors

We now prove that the rows A_g and B_g are forced by the displayed supports on the relevant cone. The argument is degree-theoretic: a selected monomial supplies a lower bound, and the strict inequalities bound every competing summand from above.

6.1 The S -phase

Lemma 6.1 (First-shear selector). *If $u \in \mathcal{C}_g$, then the two gradient rows in the S phase have degree vector $A_g u$. If a carried degree vector w satisfies $w < A_g u$ componentwise, then the corresponding S output also has degree vector $A_g u$.*

Proof. The first gradient coordinate is

$$\partial_{q_1} V = 2q_1 q_2^2 + g q_1^{g-1}.$$

Its two candidate degrees are $u_1 + 2u_2$ and $(g-1)u_1$. Since $u_2/u_1 < (g-2)/2$,

$$u_1 + 2u_2 < (g-1)u_1. \tag{6.1}$$

The pure term therefore gives the first entry of $A_g u$. The second gradient coordinate is

$$\partial_{q_2} V = 2q_1^2 q_2,$$

whose degree is $2u_1 + u_2$, giving the second entry. If $w < A_g u$, adding the carried coordinates cannot change either maximum. The selected summands are nonzero because their scalar coefficients are 2 or g and $K[q, p]$ is a domain of characteristic zero. Thus the strict support gaps give exact degrees, not only upper bounds. □

Phase-labelled interpretation of the first gap. The inequality (6.1) is equivalent to

$$(g-1)u_1 - (u_1 + 2u_2) = 2 \left(\frac{g-2}{2} - \frac{u_2}{u_1} \right) u_1 > 0. \quad (\text{S-gap})$$

Thus the margin is pointwise in the ratio, rather than merely asymptotic in g . At $n = 0$, the carried p -coordinates have degree one and the two selected gradient degrees are $g-1$ and 3. For $n \geq 1$, the carried vector is exactly $v_n = A_g u_{n-1}$; it is not an independent vector that needs a new cone assumption. Once the complete-step lemma gives $u_n > u_{n-1}$, multiplication by the nonnegative matrix A_g gives

$$v_n = A_g u_{n-1} < A_g u_n. \quad (\text{S-carried})$$

The two coordinates of (S-carried) are precisely the two carried terms in the S phase. This is why the induction is ordered as

$$u_n \in \mathcal{C}_g \implies S \text{ selector and } v_{n+1} = A_g u_n \implies T \text{ selector and } u_{n+1} = C_g u_n \implies u_{n+1} > u_n.$$

No conclusion about v_{n+1} is used before the S selector has been established.

6.2 The T -phase

Lemma 6.2 (Second-shear selector). *If $u \in \mathcal{C}_g$ and $v = A_g u$, then the gradient contribution in the T -phase has degree vector $B_g v$, and it strictly dominates the carried q -coordinates.*

Proof. The first p -gradient coordinate is

$$\partial_{p_1} W = 2p_1 p_2^2,$$

so its degree is $v_1 + 2v_2$. It dominates the carried q_1 -degree because

$$v_1 + 2v_2 - u_1 = (g-1)u_1 + 2(2u_1 + u_2) - u_1 = (g+2)u_1 + 2u_2 > 0. \quad (6.2)$$

For the second row,

$$\partial_{p_2} W = 2p_1^2 p_2 + g p_2^{g-1}.$$

The cross term has degree $2v_1 + v_2$, while the pure term has degree $(g-1)v_2$. If $r = u_2/u_1$, then

$$v_1 = (g-1)u_1, \quad v_2 = (2+r)u_1,$$

and therefore

$$\frac{v_2}{v_1} = \frac{2+r}{g-1} \geq \frac{3}{g-1} > \frac{2}{g-2},$$

where the final inequality is equivalent to $g > 4$. Hence

$$2v_1 + v_2 < (g-1)v_2. \quad (6.3)$$

The pure term selects the second row. It also dominates the carried q_2 -degree because

$$(g-1)v_2 - u_2 = 2(g-1)u_1 + (g-2)u_2 > 0. \quad (6.4)$$

□

The intermediate phase has a different ratio. Writing $r = u_2/u_1$ gives

$$v_1 = (g-1)u_1, \quad v_2 = (2+r)u_1, \quad s := \frac{v_2}{v_1} = \frac{2+r}{g-1}.$$

For $1 \leq r < R = (g-2)/2$,

$$\frac{3}{g-1} \leq s < \frac{g+2}{2(g-1)}.$$

The lower endpoint leaves the explicit margin

$$\frac{3}{g-1} - \frac{2}{g-2} = \frac{g-4}{(g-1)(g-2)} > 0. \quad (\text{T-face-margin})$$

Consequently the mixed monomial in the second T row is strictly below the pure monomial for every r in the half-open cone. The carried-coordinate gaps can be collected in the following phase ledger:

output	selected degree	carried degree	strict gap	
T_1	$q_1\text{-phase } v_1 + 2v_2$	u_1	$(g+2)u_1 + 2u_2$	(Phase gaps)
T_2	pure $(g-1)v_2$	u_2	$2(g-1)u_1 + (g-2)u_2$	
T_2	pure $(g-1)v_2$	mixed $2v_1 + v_2$	$(g-1)v_2 - (2v_1 + v_2)$	

The last entry is positive by (6.3); the first two are positive because $u_1, u_2 > 0$. Table [Phase gaps](#) is a list of three different comparisons, not three descriptions of one maximum.

6.3 The complete matrix and the half-step warning

Combining the selected rows gives

$$B_g A_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix} \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix} = C_g. \quad (6.5)$$

The phase recurrence is

$$v_{n+1} = A_g u_n, \quad u_{n+1} = B_g v_{n+1} = C_g u_n. \quad (6.6)$$

One can instead stack the two phase vectors and write a formal off-diagonal matrix, but that matrix advances only one shear at a time. Its nonzero spectrum must be interpreted with a two-phase period correction. Equation (6.5) is the complete-step theorem matrix.

7 Two-stage cone invariance

The selectors above are conditional on a cone. We now show that the complete matrix returns the q -degree vector to that same cone for every iterate.

Lemma 7.1 (Ratio map). *Let*

$$C_g = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix}.$$

If $u \in \mathcal{C}_g$, then $C_g u \in \mathcal{C}_g$ and $C_g u > u$ componentwise.

Proof. Write $r = u_2/u_1$ and $R = (g - 2)/2$. The new ratio is

$$f_g(r) = \frac{(C_g u)_2}{(C_g u)_1} = \frac{(g - 1)(2 + r)}{g + 3 + 2r}.$$

Its derivative is

$$f'_g(r) = \frac{(g - 1)^2}{(g + 3 + 2r)^2} > 0.$$

At the lower endpoint,

$$f_g(1) = \frac{3(g - 1)}{g + 5} > 1$$

for $g > 4$. At the upper endpoint,

$$f_g(R) = \frac{(g - 1)(g + 2)}{2(2g + 1)}.$$

After multiplying by the positive denominator, the inequality $f_g(R) < R$ is equivalent to

$$(g - 1)(g + 2) < (g - 2)(2g + 1),$$

whose right-minus-left side is $g(g - 4) > 0$. Monotonicity therefore gives $1 \leq f_g(r) < R$ whenever $1 \leq r < R$. Finally,

$$\begin{aligned} (C_g u)_1 - u_1 &= (g + 2)u_1 + 2u_2 > 0, \\ (C_g u)_2 - u_2 &= 2(g - 1)u_1 + (g - 2)u_2 > 0. \end{aligned}$$

This proves cone invariance and strict componentwise growth. $\square$

Endpoint margins and the half-open boundary. The endpoint computations can be written with their margins exposed:

$$f_g(1) - 1 = \frac{2(g - 4)}{g + 5}, \quad R - f_g(R) = \frac{g(g - 4)}{2(2g + 1)}. \quad (\text{Cone margins})$$

Both quantities are strictly positive exactly in the range needed here, $g \geq 5$. The upper endpoint is excluded from $\mathcal{C}_g$ because the S -phase gap vanishes there:

$$(g - 1)u_1 - (u_1 + 2Ru_1) = 0.$$

The lower endpoint is included because $f_g(1) > 1$, so the next ratio moves into the interior. Thus the induction never needs to decide which face wins at $r = R$, and it never loses the lower bound $r \geq 1$ after one step.

The cone also gives a useful two-phase picture. The map $r \mapsto f_g(r)$ acts only after both shears have been applied. During the intermediate S -phase the ratio is instead

$$r \mapsto s_g(r) = \frac{2 + r}{g - 1},$$

and the T selector uses the inequality $s_g(r) > 2/(g - 2)$ from ([T-face-margin](#)). The two maps have different fixed points and different boundary tests; treating them as one ratio map would conflate a half-step with a complete iterate.

Remark 7.2 (Why the cone has two phases). It is tempting to write one strict cone in which both $A_g u > v$ and $B_g v > u$ hold. That is not the phase structure of the map. The intermediate vector is exactly $v_{n+1} = A_g u_n$; it is not a variable that should be compared strictly with itself. The correct induction has a q -phase cone, followed by the S -selector, followed by the T -selector, and then a complete-step return through C_g .

7.1 Old-term dominance and the simultaneous phase induction

There is one carried vector that must be identified explicitly. Since the map T fixes the p -coordinates, the final p -degree vector of F_g^n is v_n for every $n \geq 1$; at $n = 0$ this remains true by the convention $v_0 = (1, 1)^\top$. Thus v_n is the old input to the next S phase, whereas v_{n+1} is the new output of that phase.

A simple coefficient fact will be used inside the induction. The coordinate polynomials of F_g have nonnegative integer coefficients, and composition preserves the semiring $\mathbb{Z}_{\geq 0}[q_1, q_2, p_1, p_2]$. Hence a strictly highest-degree summand cannot be cancelled. Lemma 8.1 records the complete coefficient-path version after the degree induction. Here we use only this semiring closure to pass from a strict support maximum to an exact degree.

Proposition 7.3 (Simultaneous carried-term induction). *For every $n \geq 0$, $u_n \in \mathcal{C}_g$, the two phase equations*

$$v_{n+1} = A_g u_n, \quad u_{n+1} = B_g v_{n+1} = C_g u_n$$

hold, and $u_{n+1} > u_n$ componentwise. At the same step all carried and competing terms satisfy

$$\max\{v_{1,n}, u_{1,n} + 2u_{2,n}\} < (g-1)u_{1,n}, \quad v_{2,n} < 2u_{1,n} + u_{2,n}, \quad (7.1)$$

$$u_{1,n} < v_{1,n+1} + 2v_{2,n+1}, \quad \max\{u_{2,n}, 2v_{1,n+1} + v_{2,n+1}\} < (g-1)v_{2,n+1}. \quad (7.2)$$

Consequently, for $n \geq 1$, $v_n = A_g u_{n-1}$.

Proof. We induct on the complete-step index. At $n = 0$ both carried vectors are $(1, 1)^\top$. The three gaps in the S phase are

$$(g-1) - (1+2) = g-4 > 0, \quad (g-1) - 1 = g-2 > 0, \quad (2+1) - 1 = 2 > 0.$$

They compare, respectively, the pure first-row term with the mixed first-row term, the pure first-row term with carried p_1 , and the second-row mixed term with carried p_2 . Hence

$$v_1 = A_g u_0 = \begin{pmatrix} g-1 \\ 3 \end{pmatrix}.$$

For the T phase the first selected term exceeds carried q_1 by

$$v_{1,1} + 2v_{2,1} - u_{1,0} = g + 4 > 0.$$

In the second row, the pure term exceeds the mixed term and carried q_2 by

$$(g-1)v_{2,1} - (2v_{1,1} + v_{2,1}) = g-4 > 0, \quad (g-1)v_{2,1} - u_{2,0} = 3g-4 > 0.$$

Thus $u_1 = B_g v_1 = C_g u_0$. Lemma 7.1 puts u_1 in $\mathcal{C}_g$ and gives $u_1 > u_0$.

Now let $n \geq 1$ and suppose the assertions have been established through step $n-1$. Put

$$\Delta_n = u_n - u_{n-1} \in \mathbb{R}_{>0}^2, \quad r_n = \frac{u_{2,n}}{u_{1,n}}.$$

The old p vector is $v_n = A_g u_{n-1}$. The first S -face gap is

$$(g-1)u_{1,n} - (u_{1,n} + 2u_{2,n}) = u_{1,n}(g-2-2r_n) > 0,$$

because $u_n \in \mathcal{C}_g$. The two carried-coordinate gaps are the entries of

$$A_g u_n - v_n = A_g \Delta_n = \begin{pmatrix} (g-1)\Delta_{1,n} \\ 2\Delta_{1,n} + \Delta_{2,n} \end{pmatrix} > 0.$$

This proves every inequality in (7.1) and gives $v_{n+1} = A_g u_n$.

Write $u = u_n$ and $v = v_{n+1} = A_g u$, so $v_1 = (g-1)u_1$ and $v_2 = 2u_1 + u_2$. The carried gap in the first T row is

$$v_1 + 2v_2 - u_1 = (g+2)u_1 + 2u_2 > 0.$$

The pure-versus-mixed gap in the second row can be cleared without a ratio denominator:

$$\begin{aligned} (g-1)v_2 - (2v_1 + v_2) &= (g-2)v_2 - 2v_1 \\ &= (g-2)u_2 - 2u_1 \\ &= u_1((g-2)r_n - 2) \geq (g-4)u_1 > 0. \end{aligned}$$

Finally, the pure second-row term exceeds carried q_2 by

$$(g-1)v_2 - u_2 = 2(g-1)u_1 + (g-2)u_2 > 0.$$

These are all inequalities in (7.2); hence $u_{n+1} = B_g v_{n+1} = C_g u_n$. Lemma 7.1 returns u_{n+1} to $\mathcal{C}_g$ and gives $u_{n+1} > u_n$, closing the induction. At each of the strict gaps, the selected summand is a nonzero scalar (2 or g) times a product of nonzero coordinate polynomials in the domain $K[q, p]$. The strict-gap conclusion therefore supplies an exact degree before the coefficient-path lemma is invoked; the latter will certify that this nonzero leading piece survives every subsequent composition. $\square$

8 No cancellation, exact recurrence, and visibility

8.1 Coefficientwise no-cancellation

Lemma 8.1 (Positive coefficients prevent cancellation). *For every n , every coordinate polynomial of F_g^n has coefficients in $\mathbb{Z}_{\geq 0}$. Every monomial selected in (6.6) has a positive coefficient, and its degree cannot be removed by cancellation.*

Proof. Perform the iteration first over the polynomial semiring

$$\mathbb{Z}_{\geq 0}[q_1, q_2, p_1, p_2]$$

and then apply the coefficient map $\mathbb{Z} \rightarrow K$. At time zero each coordinate is a single variable with coefficient one. If all four coordinates at time n lie in this semiring, then the formulas (3.1)–(3.3) form the next four coordinates using only addition, multiplication, and multiplication by 2 or g . These operations preserve the semiring, so induction proves coefficientwise nonnegativity for all iterates.

For completeness, consider the selected top-degree pieces. If $P^{[d]}$ denotes the degree- d homogeneous part of a polynomial P , then the four selected contributions have top parts of the form

$$\begin{aligned} g(q_{1,n}^{[u_{1,n}]})^{g-1}, & \quad 2(q_{1,n}^{[u_{1,n}]} q_{2,n}^{[u_{2,n}]}), \\ 2\hat{p}_{1,n+1}^{[v_{1,n+1}]} (\hat{p}_{2,n+1}^{[v_{2,n+1}]})^2, & \quad g(\hat{p}_{2,n+1}^{[v_{2,n+1}]})^{g-1}. \end{aligned}$$

Every factor displayed here is nonzero. Products of nonzero polynomials are nonzero, and all their coefficients are nonnegative integers; multiplication by 2 or g produces at least one positive integer coefficient. Proposition 7.3 places every other top-level summand at a strictly smaller total degree, so none can alter the displayed homogeneous part.

Equivalently, fully expanding an iterate expresses each monomial coefficient as a finite sum over expression-tree paths. A path weight is a product of copies of 2, g , and positive multinomial coefficients. A selected term has at least one such positive-weight path. If two paths end at the same monomial, their weights add; path uniqueness is neither asserted nor needed. Because $\text{char } K = 0$, the homomorphism $\mathbb{Z} \rightarrow K$ is injective, so no positive integer path sum becomes zero after base change. Thus every selected leading term survives and no cancellation changes any of the degrees in (6.6). $\square$

Proposition 8.2 (Leading-form propagation). *The selector inequalities determine not only the degrees but also the top homogeneous forms carried by the two phases. For $n \geq 0$, write*

$$Q_{i,n} := (q_{i,n})^{[u_{i,n}]}, \quad R_{i,n+1} := (\hat{p}_{i,n+1})^{[v_{i,n+1}]},$$

where $\hat{p}_{i,n+1} = p_{i,n} + \partial_{q_i} V(q_n)$ and $P^{[d]}$ denotes the homogeneous component of total degree d . Then

$$R_{1,n+1} = gQ_{1,n}^{g-1}, \quad R_{2,n+1} = 2Q_{1,n}^2 Q_{2,n}, \quad (8.1)$$

$$Q_{1,n+1} = 2R_{1,n+1}R_{2,n+1}^2, \quad Q_{2,n+1} = gR_{2,n+1}^{g-1}. \quad (8.2)$$

Equivalently, the complete-step top forms obey

$$Q_{1,n+1} = 8g Q_{1,n}^{g+3} Q_{2,n}^2, \quad (8.3)$$

$$Q_{2,n+1} = g2^{g-1} Q_{1,n}^{2(g-1)} Q_{2,n}^{g-1}. \quad (8.4)$$

Proof. By Proposition 7.3, the pure q_1^{g-1} term is the unique top-degree contribution to the first S output, while the mixed $q_1^2 q_2$ term is the unique top-degree contribution to the second output. Taking homogeneous parts of the two identities in (3.1) therefore gives (8.1). The same proposition says that the mixed term in the first T output and the pure p_2^{g-1} term in the second output are respectively the unique top-degree contributions in (3.2); taking homogeneous parts gives (8.2). Substitution of the first line into the second line yields (8.3)–(8.4). At the initial step this reads

$$Q_{1,0} = q_1, \quad Q_{2,0} = q_2, \quad R_{1,1} = gq_1^{g-1}, \quad R_{2,1} = 2q_1^2 q_2,$$

so the formulas start from nonzero forms. Every subsequent form is a positive scalar times a product of nonzero forms in the domain $K[q, p]$; this gives a concrete coefficient-path witness at every iterate. $\square$

For a finite exponent set $E \subset \mathbb{N}^4$ (with coordinates ordered as (q_1, q_2, p_1, p_2)), let $aE + bE' = \{ae + be' : e \in E, e' \in E'\}$ denote its Minkowski sum. The coefficientwise positivity in Lemma 8.1 makes the support version of the leading-form recurrence exact. If $E_{i,n} = \text{supp}(Q_{i,n})$, then

$$E_{1,n+1} = (g+3)E_{1,n} + 2E_{2,n}, \quad (8.5)$$

$$E_{2,n+1} = 2(g-1)E_{1,n} + (g-1)E_{2,n}. \quad (8.6)$$

Indeed, the support of a product of positive-coefficient polynomials is the Minkowski sum of their supports; coincident exponent sums only add positive coefficients. Thus (8.5)–(8.6) are a monomial-level refinement of the degree rows of C_g , not a new genericity assumption. They also show directly that the selected degree paths remain present under every composition.

8.2 The exact complete-step recurrence

Proposition 8.3 (Two-phase recurrence). *For every $n \geq 0$,*

$$v_{n+1} = A_g u_n, \quad u_{n+1} = C_g u_n.$$

Consequently $u_n = C_g^n(1, 1)^\top$.

Proof. Lemma 6.1 identifies the exact S -phase vector. Lemma 6.2 identifies the exact T -phase vector once $v_{n+1} = A_g u_n$ is known. Lemma 7.1 keeps u_n in the selector cone, while Proposition 7.3 and Lemma 8.1 ensure that no carried term or cancellation changes a selected degree. Equation (6.6) follows, and induction from $u_0 = (1, 1)^\top$ gives the closed recurrence. $\square$

Lemma 8.4 (Visibility of the total degree). *For every $n \geq 1$, the second q -coordinate has maximal degree among all four coordinates of F_g^n . In particular,*

$$\deg(F_g^n) = e_2^\top u_n.$$

Proof. Cone invariance gives $u_{n,2} > u_{n,1}$ for $n \geq 1$. The final p -degrees are $v_n = A_g u_{n-1}$. Direct subtraction gives

$$C_g - A_g = \begin{pmatrix} 4 & 2 \\ 2g-4 & g-2 \end{pmatrix},$$

whose entries are positive for $g \geq 5$. Therefore $u_n = C_g u_{n-1} > A_g u_{n-1} = v_n$ componentwise. Both final p -coordinates are below the corresponding q -coordinates, and the second q -degree is larger than the first. The claim follows. $\square$

Visibility as a degree functional. The comparison with A_g is stronger than the statement that C_g has larger row sums. For every positive vector x ,

$$(C_g - A_g)x = \begin{pmatrix} 4x_1 + 2x_2 \\ (2g-4)x_1 + (g-2)x_2 \end{pmatrix} > 0. \quad (\text{C-minus-A})$$

Thus the two final p degrees are strictly below the corresponding q degrees at every genuine step. Cone membership gives $u_{n,2} \geq u_{n,1}$, with strict inequality for $n \geq 1$, so the maximum over the four coordinate degrees is exactly the second entry of u_n . Equivalently, the total-degree functional on the full four-coordinate ledger restricts to the linear functional $e_2^\top$ on the q -phase.

This restriction is important for the asymptotic claim. A positive matrix can have a large spectral radius that is invisible to a chosen coordinate if the initial vector lies in a complementary invariant subspace or if the output functional annihilates the Perron eigenvector. Equation (C-minus-A) rules out the second failure for the actual degree ledger, while positivity of C_g rules out the first.

9 Perron calculation and dynamical degree

The recurrence is now exact, so the asymptotic calculation is a finite matrix computation with an explicitly visible coordinate.

Lemma 9.1 (Perron root). *The eigenvalues of C_g are*

$$\lambda_{\pm} = g + 1 \pm 2\sqrt{g} = (\sqrt{g} \pm 1)^2.$$

The larger root is the Perron root.

Proof. The trace and determinant are

$$\mathrm{tr}(C_g) = 2g + 2, \quad \det(C_g) = (g + 3)(g - 1) - 4(g - 1) = (g - 1)^2.$$

Thus the characteristic discriminant is

$$(2g + 2)^2 - 4(g - 1)^2 = 16g,$$

which gives the displayed roots. All four entries of C_g are positive, so the larger eigenvalue is simple and has a positive eigenvector by the Perron–Frobenius theorem. $\square$

Perron accessibility and the closed recurrence. The right Perron eigenvector can be chosen explicitly. Writing $s = \sqrt{g}$, one checks

$$C_g \begin{pmatrix} 1 \\ s - 1 \end{pmatrix} = (s + 1)^2 \begin{pmatrix} 1 \\ s - 1 \end{pmatrix}. \quad (\text{Perron vector})$$

Both entries are positive for $g \geq 5$. The corresponding left eigenvector can also be written without an existence argument:

$$\ell_+ = (\sqrt{g} + 1, 1)^\top, \quad \ell_+^\top C_g = \lambda_+ \ell_+^\top.$$

Indeed,

$$\ell_+^\top (1, 1)^\top = \sqrt{g} + 2 > 0, \quad e_2^\top (1, \sqrt{g} - 1)^\top = \sqrt{g} - 1 > 0.$$

These are the accessibility and visibility pairings needed to turn the matrix radius into the degree limit. In particular, no coordinate projection in the proof is orthogonal to the Perron class.

The same point can be verified without invoking an abstract decomposition. Cayley–Hamilton gives

$$C_g^{n+2} = 2(g + 1)C_g^{n+1} - (g - 1)^2 C_g^n.$$

Applying $e_2^\top$ on the left and $(1, 1)^\top$ on the right produces the scalar recurrence in Proposition 9.2. Its leading coefficient is

$$a_+ = \frac{3(g - 1) - \lambda_-}{\lambda_+ - \lambda_-} = \frac{(\sqrt{g} - 1)(\sqrt{g} + 2)}{2\sqrt{g}} > 0. \quad (\text{Perron coefficient})$$

Thus the λ_+^n term is present in the observed coordinate, rather than being removed by a special initial condition. This is an exact algebraic check, not a numerical estimate of the growth rate.

Proof of Theorem 3.2. Lemmas 5.1 and 5.2 give polynomial invertibility and preservation of ω . Lemmas 6.1–8.1 give the exact complete-step recurrence, and Proposition 8.3 gives $u_n = C_g^n(1, 1)^\top$. Lemma 8.4 identifies the total degree with the second component, proving (3.4). Lemma 9.1 supplies the positive root. Because both the initial vector and e_2 have positive pairing with the Perron class, the subdominant eigenvalue is exponentially negligible and

$$\lim_{n \rightarrow \infty} \left(e_2^\top C_g^n (1, 1)^\top \right)^{1/n} = \rho(C_g).$$

Finally, the pure powers q_1^{g-1} in ∇V and p_2^{g-1} in ∇W show $\deg(S) = \deg(T) = g - 1$. The strict comparison is

$$(g - 1)^2 - (\sqrt{g} + 1)^2 = \sqrt{g}(\sqrt{g} - 2)(\sqrt{g} + 1)^2 > 0$$

for $g \geq 5$. The mixed monomials join the two coordinate labels in the displayed support, which is the stated narrow coupling claim. $\square$

9.1 A scalar recurrence and an exact closed form

The matrix formula also gives a useful scalar check on the degree sequence. Set

$$d_n = e_2^\top C_g^n \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \deg(F_g^n) \quad (n \geq 1),$$

and extend the notation by $d_0 = 1$.

Proposition 9.2 (Scalar degree recurrence). *The sequence (d_n) satisfies*

$$d_{n+2} = 2(g+1)d_{n+1} - (g-1)^2 d_n, \quad d_0 = 1, \quad d_1 = 3(g-1).$$

Equivalently, with $\lambda_\pm = (\sqrt{g} \pm 1)^2$,

$$d_n = \frac{3(g-1) - \lambda_-}{\lambda_+ - \lambda_-} \lambda_+^n + \frac{\lambda_+ - 3(g-1)}{\lambda_+ - \lambda_-} \lambda_-^n.$$

Proof. The Cayley–Hamilton identity for C_g is

$$C_g^2 - 2(g+1)C_g + (g-1)^2 I = 0.$$

Multiplying on the left by $e_2^\top C_g^n$ and on the right by $(1, 1)^\top$ gives the recurrence. Since

$$C_g \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} g+5 \\ 3(g-1) \end{pmatrix},$$

the initial values follow. Writing $d_n = a\lambda_+^n + b\lambda_-^n$, the equations at $n = 0$ and $n = 1$ are $a + b = 1$ and $a\lambda_+ + b\lambda_- = 3(g-1)$. Since $\lambda_+ - \lambda_- = 4\sqrt{g}$, they give

$$a = \frac{3(g-1) - \lambda_-}{4\sqrt{g}} = \frac{(\sqrt{g}-1)(\sqrt{g}+2)}{2\sqrt{g}}, \quad b = \frac{\lambda_+ - 3(g-1)}{4\sqrt{g}} = -\frac{(\sqrt{g}+1)(\sqrt{g}-2)}{2\sqrt{g}}.$$

Substitution yields the displayed closed form and makes the signs of both coefficients explicit. $\square$

The closed form is not needed to establish the dynamical degree, but it is a useful internal consistency check: the coefficient of the Perron term is positive because C_g is positive and both e_2 and the initial vector pair positively with its Perron eigenvectors. It also makes clear why the visible-coordinate argument in Lemma 8.4 is part of the theorem rather than an after-the-fact interpretation of a matrix sequence.

9.2 Why the threshold is $g \geq 5$

The lower threshold is forced by the second selector. The strict inequality

$$\frac{3}{g-1} > \frac{2}{g-2}$$

is equivalent to $g > 4$. At $g = 4$, the endpoint comparison becomes a tie for the smallest allowed ratio, and the p_2^g face is not separated from the mixed $p_1^2 p_2$ face by the argument above. At the same time, the upper-endpoint cone calculation loses its strict margin because $g(g-4) = 0$. We therefore state the theorem only for $g \geq 5$; no continuation to $g = 4$ is hidden in the notation.

The characteristic-zero and coefficient hypotheses have the same status. The formal degree inequalities are combinatorial, but the passage from a selected summand to a nonzero leading coefficient uses the positive integers 2 and g . A sign-changing potential or a positive-characteristic specialization would require a new no-cancellation argument and is outside the present theorem.

10 The narrow non-product comparison

The phrase *non-product* has a deliberately restricted meaning. The support of V contains $q_1^2 q_2^2$, and the support of W contains $p_1^2 p_2^2$. Thus the support graph on the two labels in each phase is connected. At the degree level, this connectivity appears as the positive off-diagonal entries of C_g . If the phases were independent scalar recurrences in the displayed split, the complete transfer matrix would be diagonal after ordering the two labels; that is not what (6.5) gives.

The comparison with the elementary degree product is equally specific:

$$\lambda_1(F_g) = (\sqrt{g} + 1)^2 < (g - 1)^2 = \deg(S) \deg(T).$$

This inequality does not say that every coupled composition has submultiplicative degree growth, nor that F_g cannot be conjugated to a product in a different coordinate system. It records the exact value for this fixed word and the strict baseline comparison in the coordinates where the supports are given.

Example 10.1 (The endpoint $g = 5$: a complete hand audit). At the smallest allowed parameter,

$$A_5 = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}, \quad B_5 = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}, \quad C_5 = \begin{pmatrix} 8 & 2 \\ 8 & 4 \end{pmatrix}.$$

The cone is $1 \leq r < 3/2$. Starting with $u_0 = (1, 1)^\top$, the two phase ledgers give

$$v_1 = A_5 u_0 = (4, 3)^\top, \quad u_1 = C_5 u_0 = (10, 12)^\top,$$

and the next intermediate vector is

$$v_2 = A_5 u_1 = (40, 32)^\top, \quad u_2 = C_5 u_1 = (104, 128)^\top.$$

These values are not a numerical experiment: each entry is the selected weighted degree of a displayed monomial, and the following table records all strict comparisons at the first two complete steps.

step	phase/output	selected degree	largest competitor	gap
0	$S, \widehat{p}_1$	4	$1 + 2 = 3$	1
0	$S, \widehat{p}_2$	$2 + 1 = 3$	carry 1	2
0	$T, F_{g,1}$	$4 + 2 \cdot 3 = 10$	carry 1	9
0	$T, F_{g,2}$	$4 \cdot 3 = 12$	$\max\{2 \cdot 4 + 3, 1\} = 11$	1
1	$S, \widehat{p}_1$	$4 \cdot 10 = 40$	$\max\{10 + 24, 4\} = 34$	6
1	$S, \widehat{p}_2$	$2 \cdot 10 + 12 = 32$	carry 3	29
1	$T, F_{g,1}$	$40 + 2 \cdot 32 = 104$	carry 10	94
1	$T, F_{g,2}$	$4 \cdot 32 = 128$	$\max\{2 \cdot 40 + 32, 12\} = 112$	16

The ratio map can also be checked by hand. For

$$f_5(r) = \frac{4(2+r)}{8+2r},$$

one has $f_5(1) = 6/5$, $f_5(3/2) = 14/11 < 3/2$, and

$$f'_5(r) = \frac{16}{(8+2r)^2} > 0.$$

Thus the interval is returned strictly inside itself after one complete step. The visible degree sequence begins

$$d_0 = 1, \quad d_1 = 12, \quad d_2 = 128,$$

and Cayley–Hamilton gives

$$d_{n+2} = 12d_{n+1} - 16d_n.$$

The characteristic roots are $6 \pm 2\sqrt{5}$; the positive leading coefficient in the closed form is the concrete $g = 5$ instance of the Perron accessibility argument. In particular,

$$\lambda_1(F_5) = 6 + 2\sqrt{5} < 16 = (5 - 1)^2.$$

10.1 Support connectivity in the declared coordinate split

We make the word *coupled* precise only in the coordinate split fixed in Section 3. Let Γ_S be the undirected graph with vertices $\{q_1, q_2\}$ and an edge whenever a monomial in V contains both variables. The monomial $q_1^2 q_2^2$ gives the single edge $q_1 - q_2$. Likewise Γ_T has vertices $\{p_1, p_2\}$, and $p_1^2 p_2^2$ gives the edge $p_1 - p_2$. The two support graphs are therefore connected before any degree calculation.

The derivative ledger records how this support edge survives the shear: $\partial_{q_1} V$ contains $q_1 q_2^2$, while $\partial_{q_2} V$ contains $q_1^2 q_2$. Their degree rows are $(1, 2)$ and $(2, 1)$, so both coordinate labels occur in the first phase. The same two exponent rows occur in the mixed terms of $\partial_{p_1} W, \partial_{p_2} W$. After the two phases, the off-diagonal entries of

$$C_g = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix}$$

are exactly the algebraic record of those mixed rows. The diagonal pure-power terms are still needed to select the faces; connectivity does not mean that every monomial is selected at every phase.

This observation is deliberately weaker than a product-decomposition classification. It proves only that the displayed support cannot be split into two independent scalar ledgers without discarding a mixed monomial. It does not rule out a nonlinear change of coordinates, a different shear word, or a factorization in another category. The anti-claims in Section 11 are therefore compatible with the graph statement here.

11 Limitations, anti-claims, and collision discipline

The proof is intentionally narrow. It does not cover arbitrary sparse potentials, arbitrary coefficients, arbitrary shear words, or positive characteristic. In particular, changing signs can invalidate the coefficientwise no-cancellation step, and changing the supports can invalidate the two selected rows or the ratio cone.

We do not claim a finite Newton–fan theorem for all symplectic polynomial automorphisms, a classification in dimension four, or equality between algebraic, topological, arithmetic, and measure-theoretic entropy. We do not study periodic points, traces, multipliers, invariant curves, centralizers, torus translates, arithmetic specializations, or effective orbit counts. The paper contains no numerical or CAS certificate.

The bounded P12–P19 table in Section 2 is a collision boundary, not an exhaustive priority search. The cited literature is context only: Friedland–Milnor gives the planar baseline; Déserti and Guedj–Sibony provide higher-dimensional context; Favre–Jonsson supplies historical valuation

context; Dang–Favre motivates explicit spectral visibility; and Fujioka et al. provide a nearby four-dimensional hyperbolicity object. None of these sources proves the family, the selector inequalities, the no-cancellation statement, or the closed-form root.

The following conditions are hard stops for any revision:

1. a selector tie or an untracked carried coordinate at any iterate;
2. use of one strict cone across the phase-return equality $v_{n+1} = A_g u_n$;
3. use of a half-step matrix without a period correction;
4. cancellation made possible by signs or positive characteristic;
5. failure to show that the Perron class is reachable and visible to total degree;
6. removal of the mixed support or introduction of a universal conjugacy claim;
7. promotion of a contextual citation to proof or priority evidence.

12 Conclusion

We fixed an explicit pair of Hamiltonian shears and proved its degree growth without appealing to a generic Newton-fan principle. The proof has four interlocking parts: symmetric Hessians give canonicity and triangular inverses; the literal support monomials give A_g and B_g ; a two-stage cone plus old-term dominance and positive coefficients closes the selector induction; and the complete positive matrix C_g , together with the visible q_2 -coordinate, gives the Perron formula for the dynamical degree. The resulting value $(\sqrt{g} + 1)^2$ is strictly below the product $(g - 1)^2$ of the elementary shear degrees for every $g \geq 5$.

The conclusion is a theorem about this family and this coordinate-level comparison. Extending it to other supports or words would require a new selector and no-cancellation proof; it is not a consequence of the present argument.

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